

10. The female sex hormone is:

- (a) Estrogen (b) Androgen
- (c) Insulin (d) Oxytocin
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

Among the given options, estrogen (oestrogen) is the female sex hormone. It is produced primarily by the ovaries and responsible for the development and function of the female reproductive organs and sexual characteristics.

11. Which gland provides both endocrine and exocrine function?

- (a) Thyroid (b) Adrenal
- (c) Pancreas (d) More than one of the above
- (e) None of the above

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Ans. (c)

Pancreas is mixed gland which provide both endocrine and exocrine function. In exocrine function pancreas secretes enzymes to break down the proteins, lipids and carbohydrates in food. Functioning as an endocrine gland, the pancreas secretes the hormones insulin and glucagon to control blood sugar levels throughout the day.

VIII. Reproduction and Embryo Development

1. The entry of pollen tube into the ovule through micropyle is called

- (a) chalazogamy (b) mesogamy
- (c) anisogamy (d) porogamy

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Ans. (d)

The entry of Pollen tube into the Ovule through micropyle is called Porogamy. Mesogamy refers to a condition when Pollen Tubes enters via the integuments. Anisogamy is the form of sexual reproduction that involves the union or fusion of two gametes which differ in size and/or form.

2. Where does fertilization occur?

- (a) In uterus (b) In oviduct
- (c) In ovary (d) In vagina

45th B.P.S.C. (Pre) 2001

Ans. (b)

Fertilization occurs at the end of the fallopian tube or oviduct away from the uterus (close to the ovary). The oviduct is the tube that links ovary to the uterus and ovulated oocyte travels down to become fertilized by sperm present in the female tract.

3. Which of the functions is performed by the ovaries?

- (a) Formation of ovum (b) Secretion of progesterone
- (c) Secretion of estrogen (d) More than one of the above
- (e) None of the above

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Ans. (d)

Ovaries play a critical role in both menstruation and conception. They produce eggs for fertilization and they make the hormones estrogen and progesterone.

4. Which of the following is not a part of the female reproductive system in human beings?

- (a) Ovary (b) Uterus
- (c) Vas deferens (d) More than one of the above
- (e) None of the above

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Ans. (c)

The vas deferens are not part of the female reproductive system in humans. The vas deferens is a duct that carries sperm and is part of the male reproductive system. The female reproductive system includes Ovaries, Oviducts, Uterus, Cervix Vagina etc.

5. Asexual reproduction takes place through budding in

- (a) Amoeba (b) Yeast
- (c) Plasmodium (d) More than one of the above
- (e) None of the above

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Ans. (b)

Asexual reproduction takes place through budding in yeast. In budding, a small knob or bud forms on the parent cell grows, and then separates to become a new yeast cell. The parent cell develops tiny protrusions before developing into a bud.

PLANT PHYSIOLOGY

I. Photosynthesis

1. The stomata open or close due to change in the

- (a) position of nucleus in cells
- (b) protein composition of cells
- (c) amount of water in cells
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

The stomata are open or close due to changes in the amount of water in cells. Normally stomata are open in the day time and close during the night. The immediate cause of the opening or closing of the stomata is a change in the turgidity of the guard cell. The opening and closing of stomata will also depend on factors such as light, temperature, carbon dioxide and humidity.

2. Chlorophyll absorbs _____ wavelengths of the sunlight.
- (a) red and blue (b) green and blue
(c) green and red (d) More than one of the above
(e) None of the above

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Ans. (a)

Chlorophyll absorbs red and blue wavelengths of the sunlight and reflects green wavelengths. The absorption of blue and red wavelengths is useful for plants that have chlorophyll molecules. Plants use the energy from red and blue wavelength of the sunlight to survive and thrive.

3. Which of the following is not a component of chlorophyll?
- (a) Calcium (b) Carbon
(c) Magnesium (d) Hydrogen
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

Carbon, hydrogen, oxygen, nitrogen and magnesium are components of chlorophyll while calcium is not a component of chlorophyll.

4. The by-product of photosynthesis is :
- (a) CO₂ (b) O₂
(c) Energy (d) Sugar
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Photosynthesis occurs in membrane-bound structures called the chloroplasts. The main product of photosynthesis is glucose, which is the molecule that produces energy to run the processes of the cell. Oxygen is mainly a by-product of photosynthesis.

5. The process in which atmospheric carbon dioxide is converted into carbohydrates during photosynthesis is
- (a) oxidation (b) reduction
(c) hydrolysis (d) More than one of the above
(e) None of the above

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Ans. (b)

The process in which atmospheric carbon dioxide is converted into carbohydrates during photosynthesis is reduction.

6. Plants appear to bend towards light due to–
- (a) Concentration of auxins at shoot tip
(b) Concentration of auxins at root tip
(c) Diffusion of auxins towards shady side of shoot
(d) More than one of the above
(e) None of the above

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Ans. (c)

- The bending of plant towards light is known as phototropism. It is due to plant hormone auxins. When the growing parts of a phototropic plant detect sunlight, auxins (synthesized at the shoot tips) help the cells grow longer.
- When only one part is receiving the light, the Auxin hormone through diffusion moves to the part which is receiving less sunlight and helps in the elongation of cells that are receiving less sunlight.

7. What is the formula of glucose?

- (a) C₆H₂₂O₁₁ (b) C₆H₁₀O₅
(c) C₆H₁₂O₆ (d) C₆H₆O₆
(e) None of the above/More than one of the above

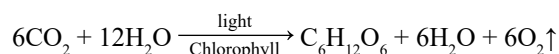
65th B.P.S.C. (Pre) 2019

48th B.P.S.C. (Pre) 2016

Ans. (c)

Photosynthesis is the process of converting light energy to chemical energy and storing it in the bonds of sugar. In this process, carbon dioxide (CO₂) is taken in from the atmosphere and oxygen (O₂) is released out during the formation of glucose (C₆H₁₂O₆).

Chemical Reaction :



8. During photosynthesis, green plants absorb :

- (a) Oxygen (b) Nitrogen
(c) Carbon monoxide (d) Carbon dioxide
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

See the explanation of the above question.

9. Which one of the following gases is essential for photosynthesis process?

- (a) CO (b) CO₂
(c) N₂ (d) O₂

43rd B.P.S.C. (Pre) 1999

Ans. (b)

See the explanation of the above question.